Sun LEH / Pythia reproduction — benchmark report

Agent-alone candidates, published prediction values and reference variants calculated in this work.

Agent Scientist: gpt-6.1-sol. Predictor names and units are listed with each measurement.

Published predictions and our predictions

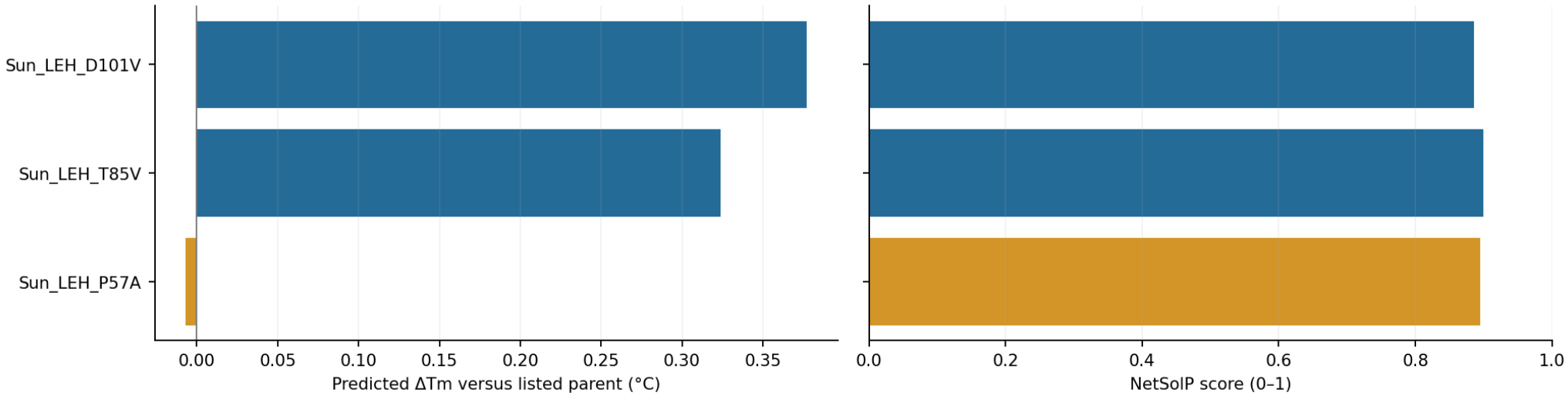
Property / predictor	Published prediction	Our candidates
Pythia score (unitless)	Candidate score not reported	D101V −12.93692160; T85V −7.93473864 (summed score)
Pythia selection cutoff	< −2	33 selected mutations using summed-score implementation
Predicted Tm change (°C)	Not reported	D101V +0.3770; T85V +0.3236

Source: Sun et al., LEH case study

Property predictions calculated in this work

Sequence	Group	Parent / background	Predicted Tm (°C)	ΔTm (°C)	NetSolP	pHPred (raw)
Sun_LEH_D101V	Our candidate	LEH_WT	52.5597	0.3770	0.886254	7.3929
Sun_LEH_T85V	Our candidate	LEH_WT	52.5063	0.3236	0.899697	7.4071
Sun_LEH_P57A	Reference	LEH_WT	52.1758	-0.0069	0.895321	7.3857

Candidate and reference property predictions



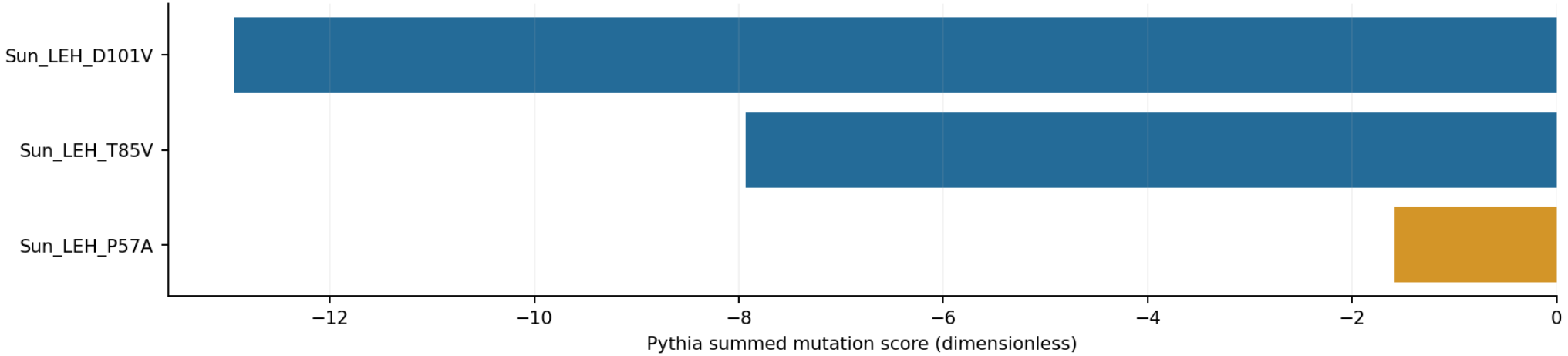
Blue: our candidates. Amber: reference variants calculated in this work. Changes use the parents listed in the table.

Pythia summed mutation score — open-source assessment

Method: Pythia C+P. Units: dimensionless. Protocol: 1NWW chain A; negative log mutant/parent ratio; fixed source and checkpoints. Values below were calculated in this work for the listed sequences.

Sequence	Group	Parent / background	Prediction (dimensionless)
Sun_LEH_D101V	Our candidate	LEH_WT	-12.9369
Sun_LEH_T85V	Our candidate	LEH_WT	-7.93474
Sun_LEH_P57A	Reported reference	LEH_WT	-1.58501

Pythia summed mutation score: reference and our sequences



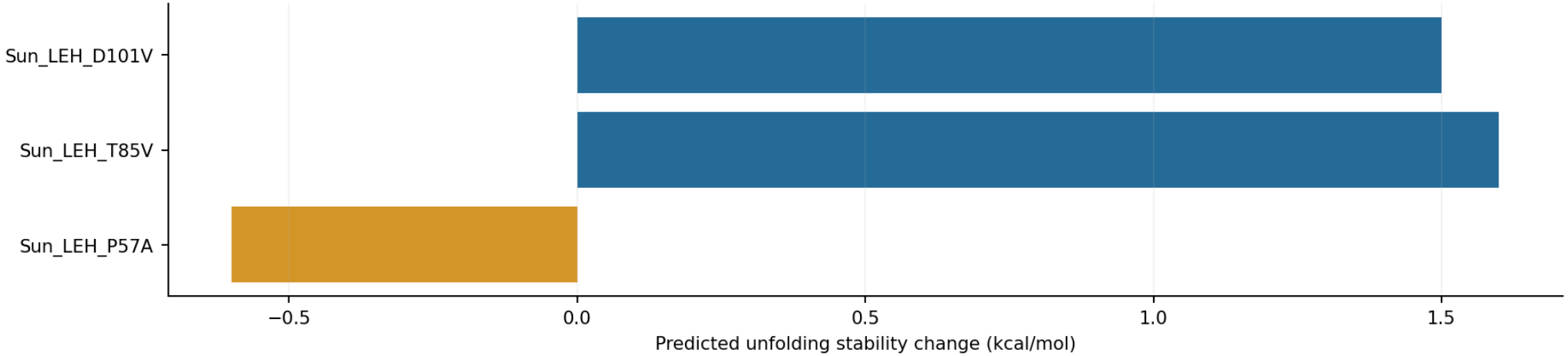
Amber: reported reference variants recalculated here. Blue: our retained candidates. All rows in this panel use the stated method and protocol.

Predicted unfolding stability change — open-source assessment

Method: DDGun SEQ. Units: kcal/mol. Protocol: sequence-based complete variant; positive predicts stabilization; compatible completed outputs reused. Values below were calculated in this work for the listed sequences.

Sequence	Group	Parent / background	Prediction (kcal/mol)
Sun_LEH_D101V	Our candidate	LEH_WT	1.5
Sun_LEH_T85V	Our candidate	LEH_WT	1.6
Sun_LEH_P57A	Reported reference	LEH_WT	-0.6

Predicted unfolding stability change: reference and our sequences



Amber: reported reference variants recalculated here. Blue: our retained candidates. All rows in this panel use the stated method and protocol.

Parent identity, preparation details and run records: results/task_005_sun_leh_pythia/agent-alone/README.md.